ML-based Wireless Localization with Differential Privacy

Locating devices using Wi-Fi signals with a formal privacy guarantee

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Introduction

Machine Learning-based **Wireless Localization** methods have become popular due to their performance and reliability for indoor **navigation**, **security**, and **communication** [2]. Several Access Points (AP) collect Channel State Information (CSI), a measurement of how Wi-Fi signals travel, and transmit it to a Central Server. The server analyzes them and predicts the device location.

However, these approaches often fail to protect training data from adversaries. They have a single point of failure – the central server. In this project, we propose an approach which accurately predicts device locations using Wi-Fi signals while guaranteeing privacy mathematically.

Threat & Privacy Concern

The neural network used to make predictions often lives on **third-party central servers**. If the server is compromised or acts maliciously, it can compromise users' training data. **Membership inference** and **model inversion** attacks can extract training information from the model [3].

Therefore, we draw the line of trust between the server and the APs. Our model architecture prevents the central server from accessing the training data while guaranteeing **Differential Privacy (DP)**.

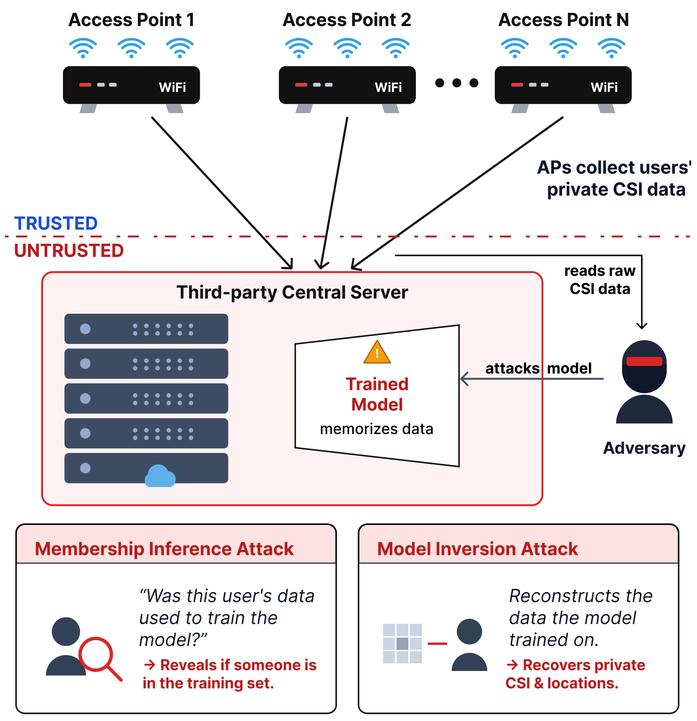


Figure 1: Privacy Threat Model: APs collect CSI and send it to an untrusted third-party server with the trained model. A compromised server lets an adversary run membership-inference and model-inversion attacks to recover the private training data.

Differential Privacy

DP is a mathematical guarantee that a model's output remains almost identical whether or not any individual's data was included in the training. A mechanism is considered (ε, δ) **-Differentially Private** if changing a single data point changes any output probability by at most a factor of e^{ε} [1].

$$\Pr[\mathcal{M}(d) \in S] \leq e^{\varepsilon} \Pr[\mathcal{M}(d') \in S] + \delta$$

In simpler terms, the **smaller the value of ε, the better the privacy** of the model. We achieve this by injecting Gaussian noise and clipping the gradients while training the model. This training algorithm is called Differentially Private Stochastic Gradient Descent (DP-SGD) [1].

Model Architecture

The model is a distributed neural network. It is split between **encoders** ($\mathbb{E}$) at the APs and a **single decoder** ($\mathbb{D}$) at the central server. Figure 2 shows the pipeline:

- **Access Point:** $\mathbb{E}_i(H_i) = \mathcal{E}_i$
- **Central Server:** $\mathbb{D}(\mathcal{E}_i) = (\sin\theta_i, \cos\theta_i)$

$$\mathbb{T}\left(\{(\sin\theta_i, \cos\theta_i)\}_{i=1}^{N_{AP}}\right) = \mathcal{L}$$

N_{AP}: Number of Access Points
H_i: CSI at the ith AP
$\mathbb{E}_i$: Encoder at i^{th} AP
$\mathcal{E}_i$: Embedding at the ith AP
$\mathbb{D}$: Decoder at the Central Server
θ_i: Angle of Arrival at the ith AP
$\mathcal{L}$: Predicted location
$\mathbb{T}$: Triangulation algorithm

We used independent ResNet-34 networks [4] for the encoders and decoder.

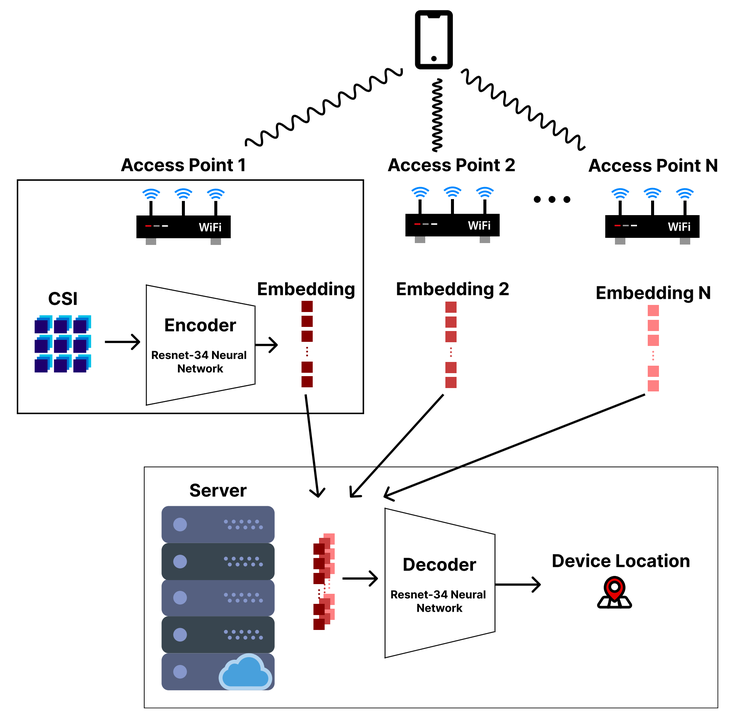


Figure 2: Model Pipeline showing Encoders, Decoders and flow of embeddings from APs to the Sever which predicts the device location

Training Algorithm

Both parts of the model are trained on the APs. Each AP contains its encoder and its local copy of the decoder.

- **Differentially Private Stochastic Gradient Descent:** DP-SGD [1] **clips per-sample gradient** individually, then adds **Gaussian noise** before each update. This bounds how much any sample can influence the model.
- **Federated Average [5]:** After every m steps, all N_{ap} APs send the weights of the decoder to the central server, which **averages** them and returns the result.
- **Dataset:** We use $jacobs_July28$ dataset containing 11,439 CSI samples collected in an $18 \times 8\ m$ indoor space with 4 Access Points.

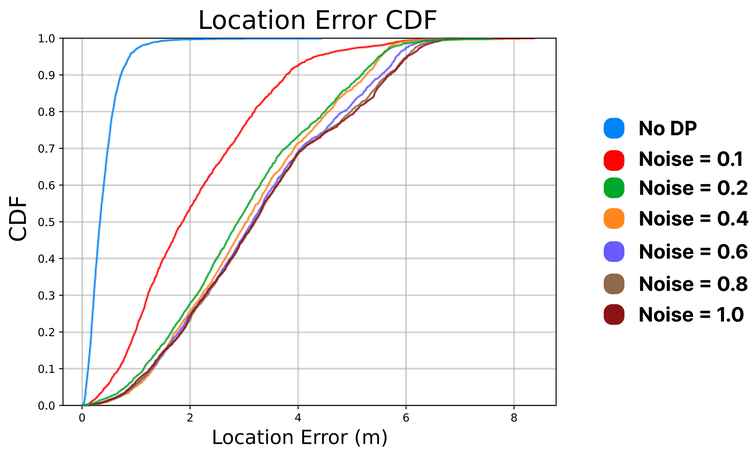


Figure 3: CDF of location error for different noise levels. The performance of the model decreases as the noise level increases and privacy claim strengthens.

Results

The model was implemented in PyTorch with Opacus. The model was trained for 25 epochs with a batch size of 32 and federated-averaged the decoder every 50 steps. Table 1 reports results across several noise levels - Gaussian noise with increasing standard deviation (σ).

Noise (σ)	ε	Location error mean (m)	Location error median (m)
0	-	0.349	0.295
0.1	28221	2.101	1.858
0.2	383.19	3.095	2.958
0.4	29.59	3.382	3.289
0.6	6.21	3.473	3.368
0.8	2.479	3.549	3.436
1.0	1.472	3.597	3.468

Table 1: Privacy cost, location error mean and median for different noise levels. Increasing noise decreases epsilon value, strengthening privacy claim, but also increases the location error mean and median.

The results show that privacy comes at a cost. Injecting noise during training degrades localization accuracy. A **lower epsilon** value **strengthens privacy** claim, **increases prediction error**. Figure 3 shows the trade-off across noise levels from Table 1.

Conclusion

The proposed distributed Split Neural Network, trained with DP-SGD and Federated Learning, provides formal (ε, δ)-Differential Privacy. This protects against membership inference attacks, keeping the training data private.

Privacy, however, comes at a cost. Stronger privacy lowers accuracy, increasing the error by almost 10 times. For most deployments, such as monitoring, navigation, and communication, this trade-off is worthwhile. In large public spaces, few meters of error rarely affects the task while privacy claim protects user data and preserves user trust.

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